

econcomplex

Technical Documentation — Version 1.0.0

Python Library for Economic Complexity
and Regional Science Indicators

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Dependencies: `numpy` ≥ 1.21 , `pandas` ≥ 1.3 , `scipy` ≥ 1.9
Compatible with Python 3.9+ (NumPy 1.x and 2.x)

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Overview

econcomplex is a Python library that consolidates the best implementations of Economic Complexity and Regional Science indicators, combining functionalities from four reference packages in the literature:

- **EconGeo** (Balland, 2017) — R package focused on geographic and regional specialization indicators
- **economiccomplexity** (Vargas Sepúlveda, 2020) — R package with C++ backend (Armadillo) for complexity indicators
- **py-econcomplexity** (The Growth Lab at Harvard, 2020) — complete Python pipeline with time series support
- **py-economic-complexity** — modular Python implementation with Polars support and cross-space indicators

Installation

```
1 # Directly from GitHub (recommended)
2 pip install git+https://github.com/eltonfreitas/econcomplex.git
3
4 # Or, for local development, from the library root directory
5 pip install -e .
6
7 # Verify the installation
8 python3 -c "import econcomplex as ec; print(ec.__version__)"
```

Listing 1: Installation via pip

Module Structure

Module	Contents
core	RCA, RPOP, Mcp, diversity, ubiquity, <code>trim_core</code> , <code>make_sample_data</code> , utilities
complexity	ECI/PCI — single entry point <code>eci_pci(method=...)</code> (eigenvector, reflections, fitness), subnational ECI
relatedness	Proximity (discrete, continuous, cosine, correlation), density, distance, co-occurrence, cross-space
specialization	LQ, Hachman, Krugman, specialization coef., similarity
inequality	Gini, locational Gini, Hoover-Gini, HHI, Shannon entropy
productivity	PRODY, EXPY, PGI, PEII
patents	Ease of Recombination, Modular Complexity
dynamics	Growth rates, entry/exit tracking (matrix and panel APIs)
outlook	COI, COG (Complexity Outlook)
optimization	ECI Optimization (Stojkoski & Hidalgo 2026), growth targeting, strategic diffusion (Alshamsi et al. 2018)
pipeline	<code>compute_complexity()</code> function — unified pipeline

API Map: Entry Points, Advanced Implementations, and Aliases

The public API has three layers. **Entry points** are what most users need; **advanced implementations** are the method-specific functions the entry points delegate to (exposed for research use); **aliases** are alternative short names bound to the very same function (no duplicated code or computation).

Entry point	Delegates to (advanced)
<code>eci_pci(method=...)</code>	<code>eci_pci_eigenvector</code> , <code>method_of_reflections</code> , <code>fitness_complexity</code> (variants: <code>mor_region</code> , <code>mor_activities</code>)
<code>proximity(continuous=..., continuous_method=...)</code>	<code>continuous_proximity</code> ; <code>cosine_proximity</code> , <code>correlation_proximity</code> , <code>density_proximity</code> , <code>distance_proximity</code> , <code>coi_proximity</code> , <code>cog_proximity</code>
<code>compute_complexity()</code>	full per-period pipeline over <code>eci_pci</code> , <code>proximity</code> , <code>density</code> , <code>COI/COG</code>

Alias	Canonical function (same object)
density, relatedness	relatedness_density
hhi	herfindahl
coi / cog	complexity_outlook_index / _gain
pgi / peii	product_gini_index / product_emissions_index
spec_coefficient	specialization_coefficient
cross_space_proximity	cross_proximity
location_quotient	rca (identical formula)

Data Organisation

This section describes the input format expected by the library, ensuring that data is correctly structured before running any indicator.

Standard Format: Long Table (*long format*)

The **econcomplex** library works exclusively with **tables in long format** (*tidy data*), where each row represents a **unique combination** of region $\times$ activity (and period, when applicable):

region	activity	value	year (optional)
SP	cnae_10	12 345	2022
SP	cnae_25	6 789	2022
RJ	cnae_10	9 012	2022
RJ	cnae_25	3 456	2022
SP	cnae_10	13 200	2023
⋮	⋮	⋮	⋮

Warning

The library does **not** accept pivoted matrices (regions in rows, activities in columns). Use `pivot_to_matrix()` only for low-level functions that explicitly require the matrix.

Data Quality Rules

1. **No duplicate rows:** each (*region*, *activity*) pair must appear exactly once per period.
2. **Non-negative values:** the value column must contain only numbers ≥ 0 .
3. **No NaN:** remove or impute missing values before passing the DataFrame.
4. **Consistent granularity:** all regions must use the same geographic level (e.g., all municipalities *or* all states).
5. **Homogeneous activities:** use a single classification per analysis (e.g., CNAE 2-digit *or* CNAE 4-digit, not mixed).

Column Names — cols Dictionary

The pipeline uses a `cols` dictionary to map the actual column names of your DataFrame:

Key	Type	Description
"loc"	required	Region / location column
"act"	required	Activity / product / technology column
"val"	required	Numeric value column (jobs, exports ...)
"time"	optional	Period column (year, quarter ...)

```

1 import pandas as pd
2 import econcomplex as ec
3
4 df = pd.read_csv("my_data.csv")
5
6 # Check structure
7 print(df.dtypes)
8 print(df.shape)
9 print(df.isnull().sum())          # check for NaN
10 print(df.duplicated().sum())      # check for duplicates
11
12 # Basic cleaning
13 df = df.dropna(subset=["region", "cnae", "jobs"])
14 df = df[df["jobs"] >= 0]          # remove negative values
15 df = df.drop_duplicates(subset=["region", "cnae", "year"])
16
17 # Run the pipeline
18 result = ec.compute_complexity(
19     df,
20     cols={"loc": "region", "act": "cnae",
21           "val": "jobs", "time": "year"},
22 )

```

Listing 2: Verifying and preparing the DataFrame before the pipeline

Supported File Formats

Format	pandas reader	Notes
.csv	pd.read_csv()	Most common; use <code>sep=";"</code> for Brazilian CSVs
.parquet	pd.read_parquet()	Ideal for large datasets (RAIS, COMEX)
.xlsx	pd.read_excel()	Requires <code>openpyxl</code> ; avoid >100 000 rows
.feather	pd.read_feather()	High performance; requires <code>pyarrow</code>
.dta	pd.read_stata()	Stata files; preserves labels

```

1 import pandas as pd
2
3 # CSV with semicolon separator (Brazilian standard)
4 df = pd.read_csv("rais.csv", sep=";", encoding="latin-1",
5                 usecols=["municipality", "cnae2d", "year", "jobs"],
6                 dtype={"municipality": str, "cnae2d": str})
7

```

```

8 # Parquet (preferred for RAIS/COMEX)
9 df = pd.read_parquet("comex_ncm.parquet",
10                     columns=["co_mun", "co_ncm", "year", "vl_fob"])
11
12 # Excel
13 df = pd.read_excel("data.xlsx", sheet_name="Sheet1",
14                   usecols=["state", "sector", "year", "jobs"])

```

Listing 3: Reading different file formats

Examples of Compatible Datasets

RAIS/MTE

Formal employment by municipality $\times$ CNAE (2 or 4 digits) $\times$ year. Value column: active employment bonds on 31/12.

COMEX/MDIC

Exports by municipality $\times$ NCM $\times$ year. Value column: VL_FOB (value in USD).

INPI — Patents

Patent applications by State of the Federation $\times$ IPC classification $\times$ year.

Atlas of Economic Complexity (Harvard)

Exports by country $\times$ product (HS) $\times$ year. Available at <https://atlas.cid.harvard.edu/>.

Simulated data (testing)

Use `ec.make_sample_data(n_locs=20, nActs=50, seed=42)` to generate a synthetic DataFrame.

How to Run the Scripts

This section describes all the ways to run the **econcomplex** library, from direct execution of Python scripts to use in interactive environments such as Jupyter and IDEs.

Prerequisites

Warning

Make sure the dependencies are installed before running any script.

```

1 # Install the library (dependencies come automatically:
2 # numpy>=1.21, pandas>=1.3, scipy>=1.9)
3 pip install git+https://github.com/eltonfreitas/econcomplex.git
4
5 # Or in development mode
6 pip install -e /path/to/econcomplex
7
8 # Verify installation
9 python3 -c "import econcomplex as ec; print(ec.__version__)"

```

Listing 4: Installing the dependencies

Direct Execution via Terminal

The simplest way to use the library is to run a `.py` script directly in the terminal:

```
1 # Navigate to the library folder
2 cd /path/to/econcomplex
3
4 # Run the full example script (guided tour of all indicators)
5 python3 examples/basic_usage.py
6
7 # Run your own script
8 python3 my_script.py
```

Listing 5: Running scripts in the terminal

If the library is **not** installed via `pip`, add the path manually at the beginning of each script:

```
1 import sys
2 sys.path.insert(0, "/path/to/econcomplex")
3
4 import econcomplex as ec
5 print(ec.__version__) # 1.0.0
```

Listing 6: Adding path to `sys.path` (without `pip` install)

Running in Jupyter Notebook / JupyterLab

```
1 cd /path/to/my_project
2 jupyter notebook
3 # or
4 jupyter lab
```

Listing 7: Opening Jupyter from your project folder

With the library installed via `pip` the first cell is just the imports (use the `sys.path` fallback only if you have not installed it):

```
1 # Cell 1 -- imports
2 import econcomplex as ec
3 import numpy as np
4 import pandas as pd
5
6 print("econcomplex", ec.__version__, "loaded successfully.")
```

Listing 8: Configuration cell in Jupyter

```
1 # Cell 2 -- load data
2 df = pd.read_csv("my_data.csv") # columns: region, sector, jobs
3
4 # Cell 3 -- compute complexity
5 result = ec.compute_complexity(
6     df,
7     cols={"loc": "region", "act": "sector", "val": "jobs"},
8     method="eigenvector",
9     compute_coi_cog=True,
10 )
11 result.head()
12
```

```

13 # Cell 4 -- display ECI by region
14 eci = (result[["region", "eci"]]
15         .drop_duplicates()
16         .set_index("region")
17         .sort_values("eci", ascending=False))
18 print(eci)

```

Listing 9: Subsequent cells — normal usage

Running in VS Code / PyCharm / Spyder

In any IDE, configure the **Python interpreter** and the **working directory** to point to the scripts folder.

```

1 // .vscode/settings.json
2 {
3     "python.defaultInterpreterPath":
4         "/Library/Frameworks/Python.framework/Versions/3.12/bin/python3",
5     "terminal.integrated.cwd": "${workspaceFolder}"
6 }

```

Listing 10: Configure path in VS Code (settings.json)

In **Spyder**, go to *Tools* → *PYTHONPATH Manager* and add the `econcomplex/` directory.

Minimal Execution Script

Ready-to-use template for any CSV-format dataset:

```

1 """
2 Usage template for the econcomplex library.
3 Adapt the column names and the CSV file path.
4 """
5 import pandas as pd
6 import econcomplex as ec
7
8 # 1. Load data
9 df = pd.read_csv("data.csv")
10 print("Columns:", df.columns.tolist())
11 print("Shape:", df.shape)
12
13 # 2. Compute all indicators
14 result = ec.compute_complexity(
15     df,
16     cols={
17         "loc": "region_column_name",      # e.g.: "municipality", "state"
18         "act": "activity_column_name",    # e.g.: "cnae", "product"
19         "val": "value_column_name",       # e.g.: "jobs", "exports"
20         "time": "year_column_name",       # optional
21     },
22     method="eigenvector",                # 'eigenvector' | 'reflections' | 'fitness'
23     threshold=1.0,
24     compute_coi_cog=True,
25 )
26
27 # 3. Export results
28 result.to_csv("complexity_results.csv", index=False)

```

```

29 print("Result saved to complexity_results.csv")
30 print("Generated columns:", sorted(result.columns.tolist()))
31
32 # 4. Individual indicators (optional)
33 mat = ec.pivot_to_matrix(df,
34                           "region_column_name",
35                           "activity_column_name",
36                           "value_column_name")
37
38 eci, pci = ec.eci_pci(mat)
39 hachman = ec.hachman_index(mat)
40 krugman = ec.krugman_index(mat)
41 entropy = ec.shannon_entropy(mat)
42
43 output = pd.DataFrame({
44     "eci": eci, "hachman": hachman,
45     "krugman": krugman, "entropy": entropy,
46 })
47 print(output.round(3))

```

Listing 11: Minimal template — copy and adapt

Running with RAIS / CNAE Data (real use case)

```

1 import pandas as pd
2 import econcomplex as ec
3
4 # Load aggregated RAIS
5 # Expected columns: municipality, cnae2d, year, jobs
6 rais = pd.read_parquet("rais_municipality_cnae.parquet")
7
8 # Filter a single year for analysis
9 rais_2022 = rais[rais["year"] == 2022].copy()
10
11 # Pipeline
12 result = ec.compute_complexity(
13     rais_2022,
14     cols={"loc": "municipality", "act": "cnae2d", "val": "jobs"},
15     method="eigenvector",
16     compute_coi_cog=False, # disable COG for large datasets
17 )
18
19 # Top 20 most complex municipalities
20 top20 = (result[["municipality", "eci"]]
21         .drop_duplicates()
22         .sort_values("eci", ascending=False)
23         .head(20))
24 print(top20.to_string(index=False))
25
26 result.to_csv("complexity_municipalities_2022.csv", index=False)

```

Listing 12: Usage with RAIS data — municipalities x CNAE

Running a Multi-Year Panel

To analyse the evolution of indicators over time, simply include the time column in the `cols` dictionary:

```

1 # rais with "year" column containing 2010, 2015, 2020
2 panel_result = ec.compute_complexity(
3     rais,
4     cols={
5         "loc": "municipality",
6         "act": "cnae2d",
7         "val": "jobs",
8         "time": "year",          # activates the temporal loop
9     },
10    method="reflections",
11    compute_coi_cog=False,
12 )
13
14 # The result already contains all years
15 print(panel_result["year"].value_counts().sort_index())
16
17 # Median ECI evolution by year
18 eci_trend = (panel_result[["year", "municipality", "eci"]]
19             .drop_duplicates()
20             .groupby("year")["eci"]
21             .median()
22             .round(4))
23 print(eci_trend)

```

Listing 13: Panel execution — multiple years

Note

The pipeline automatically iterates over each period, ensuring that the proximity matrix and complexity vectors are recalculated independently for each year.

Core Indicators (core)

Revealed Comparative Advantage (RCA)

Balassa's RCA is the main specialization filter:

$$RCA_{il} = \frac{x_{il} / \sum_l x_{il}}{\sum_i x_{il} / \sum_{il} x_{il}}$$

```

1 import econcomplex as ec
2 import pandas as pd
3
4 df = pd.read_csv("data.csv")
5 mat = ec.pivot_to_matrix(df, "region", "sector", "jobs")
6
7 rca      = ec.rca(mat)          # continuous values
8 mcp      = ec.mcp(mat, threshold=1) # 1 if RCA >= 1, 0 otherwise
9 rca_rpop = ec.rpop(mat, pop)    # Puga & Turrini RCA (requires
                                # population vector)

```

Listing 14: Continuous and binary RCA

Diversity and Ubiquity

```

1 div = ec.diversity(mat)      # number of sectors with RCA >= 1 per region
2 ubi = ec.ubiquity(mat)      # number of regions with RCA >= 1 per sector

```

Listing 15: Diversity and ubiquity

Economic Complexity (complexity)

Eigenvector Method (Hidalgo & Hausmann, 2009)

$$\text{ECI} = \frac{\vec{v}_2 - \mu(\vec{v}_2)}{\sigma(\vec{v}_2)}, \quad \text{PCI} = \frac{\vec{u}_2 - \mu(\vec{u}_2)}{\sigma(\vec{u}_2)}$$

where $\vec{v}_2$ is the second left eigenvector of matrix $\tilde{M}$:

$$\tilde{M}_{ij} = \frac{M_{ij}}{k_{c,0} \cdot k_{p,0}}$$

```

1 eci, pci = ec.eci_pci(mat, method="eigenvector")
2 print(eci.head())    # Series indexed by region
3 print(pci.head())    # Series indexed by product/sector
4
5 # Pre-processing: units with zero diversity/ubiquity are trimmed
6 # automatically and returned as NaN. Disable with trim=False, or
7 # restrict to the well-connected core with dmin=2, umin=2 (sparse data).
8 core = ec.trim_core(mat, dmin=2, umin=2) # also available standalone

```

Listing 16: ECI and PCI using the eigenvector method

Method of Reflections (Hidalgo & Hausmann, 2009)

```

1 eci_r, pci_r = ec.eci_pci(mat, method="reflections", iterations=20)
2 # tol=1e-10 stops the iterations early once the z-scored reflections
3 # stabilize; signs are oriented as in the eigenvector method
4 # (ECI correlates positively with diversity, PCI negatively
5 # with ubiquity).

```

Listing 17: Method of Reflections

Fitness-Complexity (Tacchella et al., 2012)

$$F_c^{(n+1)} = \sum_p M_{cp} Q_p^{(n)}, \quad Q_p^{(n+1)} = \frac{1}{\sum_c M_{cp} / F_c^{(n)}}$$

```

1 fit, comp = ec.eci_pci(mat, method="fitness", iterations=20)
2 # iterations is a cap (20, as in the R economiocomplexity package);
3 # the loop stops at convergence (tol) and warns if it is genuinely
4 # unstable. log_fitness=True returns the log scale recommended by
5 # Cristelli et al. (2015):
6 fit_log, comp_log = ec.fitness_complexity(mat, log_fitness=True)

```

Listing 18: Fitness-Complexity

Subnational ECI

```
1 # pci_national: PCI computed from national/international data
2 eci_sub = ec.subnational_eci(mat_sub, pci_national)
```

Listing 19: Subnational ECI with external PCI

Product Space and Relatedness (relatedness)

Proximity (ϕ)

$$\phi_{ij} = \min(P(\text{RCA}_i \geq 1 \mid \text{RCA}_j \geq 1), P(\text{RCA}_j \geq 1 \mid \text{RCA}_i \geq 1))$$

```
1 phi_d = ec.proximity(mat)["product"] # based on binary
    Mcp
2 phi_c = ec.proximity(mat, continuous=True)["product"] # based on
    continuous RCA
```

Listing 20: Discrete and continuous proximity

Density and Distance

$$\omega_{ij} = \frac{\sum_k M_{kj} \phi_{ij}}{\sum_k \phi_{ij}}, \quad d_{ij} = 1 - \omega_{ij}$$

```
1 density = ec.density(mat, phi_d) # DataFrame loc x act
2 distance = ec.distance(mat, phi_d) # DataFrame loc x act
```

Listing 21: Density and distance

Co-occurrence and Relatedness Indices

```
1 cooc = ec.co_occurrence(mat)
2 phi_cos = ec.cosine_proximity(mat) # cosine-based
3 phi_cor = ec.correlation_proximity(mat)
4 rca_rel = ec.relatedness(mat, phi_d) # relatedness densities
```

Listing 22: Co-occurrence and relatedness indices

Cross-Space (relatedness between distinct classifications)

```
1 # phi_cross: proximity between sectors and technologies (different
    matrices)
2 phi_cross = ec.cross_space_proximity(mat_sector, mat_tech)
3 density_c = ec.cross_relatedness(mat_sector, phi_cross)
```

Listing 23: Cross-space proximity and relatedness

Regional Specialization (specialization)

```

1 lq      = ec.location_quotient(mat)      # equivalent to RCA
2 hachman = ec.hachman_index(mat)          # 0 (diversified) to 1
  (specialized)
3 krugman = ec.krugman_index(mat)          # structural difference between
  regions
4 coef_spec = ec.spec_coefficient(mat)     # specialization coefficient
5 sim_exp  = ec.export_similarity(mat)     # productive structure
  similarity

```

Listing 24: Specialization indices

Inequality and Concentration (inequality)

```

1 gini_r   = ec.gini(mat)                  # Gini of value distribution
2 gini_loc = ec.locational_gini(mat)       # locational Gini (by sector)
3 hoover   = ec.hoover_gini(mat)          # Hoover index
4 hhi      = ec.hhi(mat)                  # Herfindahl-Hirschman
5 entropy  = ec.shannon_entropy(mat)       # Shannon entropy (by region)

```

Listing 25: Inequality indicators

Productivity (productivity)

```

1 prody = ec.prody(mat, gdp_per_capita)    # income associated with the
  product
2 expy  = ec.expy(mat, gdp_per_capita)     # sophistication of the export
  basket
3 pgi    = ec.pgi(mat, gini_vec)           # Product Gini Index (regional
  Gini vector)
4 peii   = ec.peii(mat, emissions)        # Product Emissions Intensity
  Index

```

Listing 26: Productivity and product inequality

Patent-Based Complexity (patents)

```

1 ease_rec = ec.ease_of_recombination(mat_pat)
2 mod_comp = ec.modular_complexity(mat_pat)

```

Listing 27: Patent indicators (input: patent x technology matrix)

Temporal Dynamics (dynamics)

```

1 growth = ec.growth_rates(df, "region", "sector", "jobs", "year")
2 entries = ec.entry_tracking(df, "region", "sector", "jobs", "year")
3 exits  = ec.exit_tracking(df, "region", "sector", "jobs", "year")

```

Listing 28: Growth rates and entry/exit tracking

Complexity Outlook (outlook)

The **COI** measures the latent diversification potential; the **COG** measures the expected complexity gain:

$$\text{COI}_c = \sum_p (1 - M_{cp}) \omega_{cp} Q_p$$

```
1 coi = ec.coi(mat, pci, phi=phi_d)
2 cog = ec.cog(mat, pci, phi=phi_d)
```

Listing 29: COI and COG

ECI Optimization (optimization)

A target-oriented optimization layer for strategic diversification (Stojkoski & Hidalgo, 2026, *Research Policy* 55, 105454). A stepping-stone forecast model is calibrated on a historical panel, the effort W_{cp} (the added RCA an economy needs to enter an activity) is obtained in closed form, and a 0–1 integer program selects the minimal-effort portfolio of new activities that reaches a target ECI.

```
1 # 1. Calibrate the stepping-stone model on a panel (t, t+5, t+10)
2 model = ec.calibrate_steppingstone(df, "region", "sector", "jobs",
3                                   "year", horizon=10, steppingstone=5)
4
5 # 2. Effort and no-policy forecast for the latest year
6 W = ec.effort_matrix(mat, model) # added RCA per candidate
7 fc = ec.forecast_specialization(mat, model) # rca, mcp, pci, eci
8
9 # 3. Minimal-effort portfolio raising each location's ECI by 0.1
10 portfolio = ec.eci_optimization(mat, model, delta_eci=0.1)
11
12 # 4. Optional: growth targeting - convert a growth target (3.5%/yr)
13 #    into an ECI target and optimize towards it
14 gm = ec.calibrate_growth_model(panel, "region", "year", "gdppc", "eci")
15 eci_star = ec.eci_target_for_growth(gm, 0.035, gdppc_now)
16 portfolio = ec.eci_optimization(mat, model, target_eci=eci_star)
```

Listing 30: ECI Optimization workflow

Strategic Diffusion (optimization.diffusion)

Complex-contagion model of diversification on the network of related activities (Alshamsi, Pinheiro & Hidalgo, 2018, *Nature Communications* 9, 1328): the probability of entering an activity is $p_i = B(\text{fraction of related activities present})^\alpha$. The greedy strategy of always entering the most related activity is suboptimal; minimal diversification time requires targeting hubs in a narrow early window.

```
1 adj = ec.proximity_network(mat, phi_threshold=0.55) # product space
2 fit = ec.calibrate_contagion(df, "region", "sector", "jobs", "year",
3                             adjacency=adj) # B and alpha
4
5 active0 = ec.mcp(mat).loc["my_region"] # current portfolio
```



```

6 seq = ec.diversification_strategy(adj, active0, B=fit["B"],
7                               alpha=fit["alpha"], strategy="greedy")
8 table = ec.compare_strategies(adj, active0, B=fit["B"],
9                               alpha=fit["alpha"])
10 best = ec.optimize_sequence(adj, active0, B=fit["B"],
11                             alpha=fit["alpha"])

```

Listing 31: Strategic diffusion workflow

Unified Pipeline (compute_complexity)

Function Signature

```

1 def compute_complexity(
2     data: pd.DataFrame,
3     cols: dict,          # {"loc": str, "act": str, "val": str,
4                          "time": str}
5     *,
6     method: str = "eigenvector", # 'eigenvector' / 'reflections' /
7     'fitness'
8     presence_test: str = "rca",   # 'rca' / 'rpop' / 'both' / 'manual'
9     threshold: float = 1.0,       # binarization threshold
10    iterations: int = 20,          # for reflections/fitness
11    proximity_type: str = "discrete", # 'discrete' / 'continuous'
12    proximity_method: str = "max",  # 'max' / 'sqrt' / 'min'
13    pop: pd.DataFrame | None = None, # population (for 'rpop'/'both')
14    compute_coi_cog: bool = True,
15    time_col: str | None = None,
16 ) -> pd.DataFrame:
17     """
18     Returns the original DataFrame enriched with columns:
19     rca, mcp, diversity, ubiquity, eci, pci,
20     density, distance, [coi, cog]
21     """

```

Listing 32: Pipeline function signature

Parameters

Parameter	Type	Description
<code>data</code>	<code>DataFrame</code>	Table in long format
<code>cols</code>	<code>dict</code>	Column mapping (see Section 2.3)
<code>method</code>	<code>str</code>	Complexity method (single entry point)
<code>presence_test</code>	<code>str</code>	How to binarize Mcp (RCA, RPOP, both, manual)
<code>threshold</code>	<code>float</code>	Binarization threshold (default: 1.0)
<code>iterations</code>	<code>int</code>	Iterations for iterative methods (default: 20)
<code>proximity_type</code>	<code>str</code>	Discrete (co-occurrence) or continuous proximity
<code>proximity_method</code>	<code>str</code>	Proximity normalization
<code>pop</code>	<code>DataFrame</code>	Population, required for RPOP
<code>compute_coi_cog</code>	<code>bool</code>	Compute COI and COG (slower)

Note

Degenerate units (zero diversity or ubiquity) are automatically trimmed by the complexity step and returned as NaN (see `trim_core`).

Complete Examples

Example 1 — Regional Employment Analysis

```

1 import pandas as pd
2 import numpy as np
3 import econcomplex as ec
4
5 # Generate synthetic data (or load a real CSV)
6 df = ec.make_sample_data(n_locs=50, n_acts=30, seed=42)
7 # Columns: loc, act, val
8
9 # Full pipeline
10 result = ec.compute_complexity(
11     df,
12     cols={"loc": "loc", "act": "act", "val": "val"},
13     method="eigenvector",
14     compute_coi_cog=True,
15 )
16
17 # Summary of generated indicators
18 print(result.columns.tolist())
19 # ['loc', 'act', 'val', 'rcá', 'mcp', 'diversity', 'ubiquity',
20 #  'éćí', 'pcí', 'density', 'distancé', 'coí', 'cog']
21
22 # Top 10 regions by ECI

```

```

23 top_eci = (result[["loc","eci"]]
24             .drop_duplicates()
25             .sort_values("eci", ascending=False)
26             .head(10))
27 print(top_eci)
28
29 # Save
30 result.to_csv("regional_result.csv", index=False)

```

Listing 33: Complete regional employment analysis

Example 2 — Exports with COMEX Data

```

1 import pandas as pd
2 import econcomplex as ec
3
4 # Load COMEX dataset aggregated by municipality x NCM x year
5 comex = pd.read_parquet("comex_mun_ncm.parquet")
6 comex_2023 = comex[comex["year"] == 2023].copy()
7
8 # Cleaning
9 comex_2023 = comex_2023.dropna(subset=["co_mun","co_ncm","vl_fob"])
10 comex_2023 = comex_2023[comex_2023["vl_fob"] > 0]
11
12 # Pipeline
13 result = ec.compute_complexity(
14     comex_2023,
15     cols={"loc": "co_mun", "act": "co_ncm", "val": "vl_fob"},
16     method="eigenvector",
17     compute_coi_cog=True,
18 )
19
20 # Save
21 result.to_csv("export_complexity_2023.csv", index=False)
22 print("Mean ECI:", result["eci"].dropna().mean().round(4))

```

Listing 34: Export analysis with COMEX/MDIC data

Example 3 — RAIS Time Panel (2010–2022)

```

1 import pandas as pd
2 import econcomplex as ec
3
4 rais = pd.read_parquet("rais_municipality_cnae_2010_2022.parquet")
5
6 # Temporal pipeline -- iterates over each year automatically
7 panel = ec.compute_complexity(
8     rais,
9     cols={
10         "loc": "municipality",
11         "act": "cnae2d",
12         "val": "jobs",
13         "time": "year",
14     },
15     method="eigenvector",
16     compute_coi_cog=False,
17 )

```

```

18
19 # ECI median evolution
20 eci_trend = (panel[["year", "municipality", "eci"]]
21              .drop_duplicates()
22              .groupby("year")["eci"]
23              .median()
24              .round(4))
25 print(eci_trend)
26
27 panel.to_csv("panel_complexity_2010_2022.csv", index=False)

```

Listing 35: Pipeline with panel data — multiple periods

Example 4 — Diversification Targets (ECI Optimization)

Step-by-step use of the optimization layer with a multi-year panel (needs at least the periods t , $t + \tau$ and $t + \Delta t$):

```

1 import pandas as pd
2 import econcomplex as ec
3
4 # 1. Panel with at least t, t+5, t+10 (e.g. 2005, 2010, 2015)
5 rais = pd.read_parquet("rais_municipality_cnae_2005_2015.parquet")
6
7 # 2. Calibrate the stepping-stone forecast model
8 model = ec.calibrate_steppingstone(
9     rais, "municipality", "cnae2d", "jobs", "year",
10     horizon=10, steppingstone=5,
11 )
12
13 # 3. Matrix of the latest year and minimal-effort portfolios
14 mat = ec.pivot_to_matrix(rais[rais["year"] == 2015],
15                           "municipality", "cnae2d", "jobs")
16 portfolio = ec.eci_optimization(mat, model, delta_eci=0.1)
17 print(portfolio.head()) # location, activity, effort, pci, targets
18
19 # 4. Optional: growth targeting (needs GDP per capita and ECI panel)
20 gm = ec.calibrate_growth_model(macro, "municipality", "year",
21                                "gdppc", "eci", horizon=10)
22 eci_star = ec.eci_target_for_growth(gm, 0.035, gdppc_now)
23 portfolio = ec.eci_optimization(mat, model, target_eci=eci_star)
24
25 # 5. Optional: diversification timing (strategic diffusion)
26 adj = ec.proximity_network(mat, phi_threshold=0.55)
27 fit = ec.calibrate_contagion(rais, "municipality", "cnae2d",
28                               "jobs", "year", adjacency=adj)
29 active0 = ec.mcp(mat).loc["my_municipality"]
30 best = ec.optimize_sequence(adj, active0,
31                             B=fit["B"], alpha=fit["alpha"])
32 print(best["sequence"][:10], best["improvement"])

```

Listing 36: ECI Optimization end to end

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```
1 import econcomplex as ec
2 print(ec.__version__)    # 1.0.0
```

Listing 37: Check the library version